

A047 · LINGO: A Knowledge Graph-Grounded Foundation Model for Single-Cell Lineage Inference

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Single-cell lineage tracing (scLT) enables reconstruction of developmental histories by coupling lineage barcodes with transcriptomic profiling. The rapid accumulation of scLT datasets across diverse species, tissues, and experimental systems has created unprecedented opportunities to uncover general principles of cell fate determination. However, lineage inference remains challenging because lineage information is often incomplete owing to barcode dropout, barcode silencing, and destructive sampling, while existing computational methods are typically tailored to individual datasets and transfer poorly across biological contexts. Meanwhile, current single-cell foundation models are largely transcriptome-driven and do not explicitly incorporate lineage-relevant biological knowledge. Here we present LINGO, a lineage foundation model that integrates a lineage-informed knowledge graph with large-scale scLT pretraining for single-cell lineage inference. LINGO combines developmental hierarchies, gene regulatory interactions, pathway information, and lineage-relevant cellular relationships within a unified graph framework and uses graph-based contrastive pretraining to learn lineage-aware representations. We pretrained LINGO on more than 2 million cells from lineage-tracing datasets spanning 18 tissue types across mouse, human, and zebrafish, and evaluated it on held-out benchmarks encompassing cell linkage prediction, lineage origin prediction, and clonal plasticity prediction across diverse lineage-tracing technologies, tissues, and species. LINGO consistently outperformed state-of-the-art lineage-analysis methods and transcriptome-based foundation models, achieving gains of up to 26 percentage points on held-out benchmarks. Furthermore, pretrained representations remained highly informative even when model weights were frozen during downstream fine-tuning, demonstrating the robustness and transferability of the learned representations. Together, these results demonstrate the value of integrating structured biological knowledge into large-scale representation learning and establish a foundation-model framework for single-cell lineage inference.

LINGO: A Knowledge Graph-Grounded Foundation Model for Single-Cell Lineage Inference

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Abstract

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